

LeafOS Model Assignment ASCII Tree Diagrams

Supporting document for model-library.md and W0-006-C

Branch: ~C Version target: 0.6.0-0.8.0

1 Purpose

This document provides black-and-white ASCII tree diagrams for the LeafOS model assignment layer.

The companion files are:

- model-library.md
- model-library-real-assignments.md
- W0-006-C-model-assignment-library-script.md
- leaf_model_assignments.py

The diagrams are intentionally terminal-readable. No color. No Unicode dependency. No font ritual. A small mercy from the machinery.

2 Core Rule

```
MODEL ASSIGNMENT IS NOT JUST A MODEL NAME.
```

```
Real assignment tuple:
```

```
model
quant
lane
endpoint
contract
parser
gate
```

```
Pipeline:
```

```
model endpoint -> logical lane -> output contract -> parser -> accepted artifact
```

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4 D-001: Model Assignment Control Tree

```
LeafOS Model Assignment Layer
|
+-- documentation/
|   |
|   +-- model-library.md
|       |
|       +-- model roles
|       +-- real assignments
|       +-- quant policy
|       +-- performance predictions
|       +-- routing doctrine
|
|   +-- model-library-real-assignments.md
|       |
|       +-- primary Brain candidates
|       +-- primary Coder assignment
|       +-- Helper assignments
|       +-- Reviewer assignment
|
+-- executable helper/
|   |
|   +-- leaf_model_assignments.py
|       |
|       +-- summary
|       +-- routes
|       +-- write-json
|       +-- doctor
|
+-- generated machine registry/
|   |
|   +-- models/model-assignments.json
|       |
|       +-- profiles[]
|       +-- routes{}
|       +-- doctrine[]
|
+-- runtime use/
|   |
|   +-- agent-brain
|   +-- agent-coder
|   +-- agent-review
|   +-- agent-report
|   +-- agent-gate
```

5 D-002: Stable Default Route

```
STABLE ROUTE
|
+-- Brain lane
|   |
|   +-- profile_id:
|       |
|       +-- brain-primary-14b-gguf
|
|   +-- model:
|       |
|       +-- tvall43/Qwen3.5-14B-A3B-Claude-4.6-Opus-Reasoning-Distilled-reap-gguf
|
|   +-- lanes:
|       |
|       +-- slot.brain.plan
|       +-- slot.brain.repair
|       +-- slot.review.architecture
|
|   +-- contracts:
|       |
|       +-- json_taskpack
|       +-- agent_graph_json
|       +-- review_json
```

```

+-- Coder lane
|
|   +-- profile_id:
|   |   coder-primary-12b-fable
|   |
|   +-- model:
|   |   yuxinlul/gemma-4-12B-coder-fable5-composer2.5-v1-GGUF
|   |
|   +-- quant:
|   |   Q4_K_M
|   |
|   +-- lane:
|   |   slot.coder.patch
|   |
|   +-- contract:
|   |   unified_diff
|
+-- Helper lane
|
|   +-- model:
|   |   same Fable 12B GGUF
|   |
|   +-- quant:
|   |   Q3_K_M
|   |
|   +-- lanes:
|   |   slot.helper.docs
|   |   slot.helper.testgen
|   |   slot.helper.repair
|
+-- Reviewer lane
|
|   +-- model:
|   |   brain-primary-14b-gguf
|   |
|   +-- contract:
|   |   review_json

```

6 D-003: Quality Experimental Route

```

QUALITY ROUTE
|
+-- Brain lane: experimental primary
|
|   +-- profile_id:
|   |   brain-primary-27b-nvfp4-mtp
|   |
|   +-- model:
|   |   Brian6145/Qwen3.6-27B-Claude-Opus-Sonnet-Distilled-NVFP4-MTP
|   |
|   +-- runtime target:
|   |   vLLM or SGLang if NVFP4/MTP support is stable
|   |
|   +-- best use:
|   |   hard architecture
|   |   graph generation
|   |   deep repair planning
|   |   completion gate design
|   |
|   +-- contract:
|   |   strict JSON taskpack / graph / repair plan
|
+-- Coder lane: careful
|
|   +-- model:
|   |   Fable 12B GGUF
|   |
|   +-- quant:
|   |   Q6_K

```

```

|
|
| +--- contract:
|     unified_diff only
|
+--- Helper lane
|
| +--- model:
|     Fable 12B GGUF
|
| +--- quant:
|     Q3_K_M
|
+--- Reviewer lane
|
| +--- model:
|     14B GGUF reasoning planner
|
| +--- contract:
|     review_json
|
QUALITY ROUTE POLICY
|
+--- if 27B runtime is stable:
|     use for hard planning
|
+--- if 27B runtime is unstable:
|     demote to experimental
|     use 14B route instead

```

7 D-004: Low-Memory Route

```

LOW-MEMORY ROUTE
|
+--- Brain
|
| +--- preferred:
|     brain-primary-14b-gguf
|
| +--- if too heavy:
|     mock route or smaller future Brain
|
| +--- contract:
|     json_taskpack
|
+--- Coder
|
| +--- model:
|     Fable 12B GGUF
|
| +--- quant:
|     Q3_K_M
|
| +--- expected behavior:
|     faster
|     lower memory
|     somewhat weaker patches
|
+--- Helper
|
| +--- model:
|     Fable 12B GGUF
|
| +--- quant:
|     Q2_K
|
| +--- use:
|     classification
|     tiny docs
|     rough repair hints

```

```

|
+-- Reviewer
|
|   +-- model:
|       helper-coder-fable-lowquant
|
|   +-- contract:
|       review_json
|
WARNING
|
+-- Q2_K is not the main coder.
+-- Q2_K is emergency helper mode.
+-- A tiny model giving confident nonsense is still nonsense.

```

8 D-005: Repair Route

```

REPAIR ROUTE
|
+-- failure detected
|   |
|   +-- parse_failure
|   +-- schema_failure
|   +-- scope_failure
|   +-- patch_apply_failure
|   +-- test_failure
|   +-- missing_completion_gate
|
+-- classify failure
|   |
|   +-- run parser logs
|   +-- inspect verify_result.json
|   +-- inspect patch_scope.json
|
+-- Brain repair plan
|   |
|   +-- model:
|       brain-primary-14b-gguf
|   |
|   +-- output:
|       repair_plan.json
|
+-- Fast repair attempt
|   |
|   +-- model:
|       Fable 12B Q3_K_M
|   |
|   +-- contract:
|       unified_diff
|
+-- if fast repair fails
|   |
|   +-- careful repair attempt
|       |
|       +-- model:
|           Fable 12B Q6_K
|       |
|       +-- contract:
|           unified_diff
|
+-- reviewer check
|   |
|   +-- model:
|       brain-primary-14b-gguf or reviewer lane
|
+-- verification
|   |
|   +-- git apply --check
|   +-- tests

```

```

+-- completion gate
+-- report

CAPS
|
+-- max repair attempts per node: 2 or 3
+-- no infinite agent hamster wheel

```

9 D-006: Brain / Coder / Helper / Reviewer Lane Split

```

LeafOS Runtime Scheduler
|
+-- slot.brain.plan
| |
| | +-- role:
| | |   Brain
| | |
| | +-- accepted contract:
| | |   json_taskpack
| | |   agent_graph_json
| | |
| | +-- parser:
| | |   parse_brain_taskpack.py
| | |
+-- slot.coder.patch
| |
| | +-- role:
| | |   Coder
| | |
| | +-- accepted contract:
| | |   unified_diff
| | |
| | +-- parser:
| | |   parse_coder_patch.sh
| | |
+-- slot.helper.docs
| |
| | +-- role:
| | |   Helper
| | |
| | +-- accepted contract:
| | |   report_markdown or unified_diff
| | |
| | +-- parser:
| | |   parse_report.sh or parse_coder_patch.sh
| | |
+-- slot.helper.testgen
| |
| | +-- role:
| | |   Helper test writer
| | |
| | +-- accepted contract:
| | |   unified_diff
| | |
+-- slot.review.scope
| |
| | +-- role:
| | |   Reviewer
| | |
| | +-- accepted contract:
| | |   review_json
| | |
| | +-- parser:
| | |   parse_review_result.py

NOTE
|
+-- lanes are scheduler roles
+-- endpoints are actual model servers

```

```

+-- parsers are the trust boundary

```

10 D-007: Quant Assignment Ladder

Fable 12B GGUF Quant Ladder	
16.0	GGUF 16-bit float
15.0	GGUF 15-bit float
14.0	GGUF 14-bit float
13.0	GGUF 13-bit float
12.0	GGUF 12-bit float
11.0	GGUF 11-bit float
10.0	GGUF 10-bit float
9.0	GGUF 9-bit float
8.0	GGUF 8-bit float
7.0	GGUF 7-bit float
6.0	GGUF 6-bit float
5.0	GGUF 5-bit float
4.0	GGUF 4-bit float
3.0	GGUF 3-bit float
2.0	GGUF 2-bit float
1.0	GGUF 1-bit float

```

+-- Q2_K
|
|   +-- size:
|   |       approx 4.83 GB
|   |
|   +-- role:
|   |       emergency helper
|   |
|   +-- use:
|   |       tiny repair hints
|   |       rough classification
|   |       low-memory fallback
|
+-- Q3_K_M
|
|   +-- size:
|   |       approx 6.09 GB
|   |
|   +-- role:
|   |       helper / fast secondary coder
|   |
|   +-- use:
|   |       docs
|   |       tests
|   |       fast repair
|
+-- Q4_K_M
|
|   +-- size:
|   |       approx 7.38 GB
|   |
|   +-- role:
|   |       default coder
|   |
|   +-- use:
|   |       normal patches
|   |       project templates
|   |       README/test/doc gates
|
+-- Q6_K
|
|   +-- size:
|   |       approx 9.79 GB
|   |
|   +-- role:
|   |       careful coder
|   |
|   +-- use:
|   |       difficult patches
|   |       repeated repair failures
|   |       quality route
|
+-- Q8_0
|
|   +-- size:
|   |       approx 12.7 GB
|   |
|   +-- role:
|   |       high-quality but VRAM-tight
|   |
|   +-- use:
|   |       only if runtime confirms fit

```



```

SUSPICIOUS TINY ARTIFACT RULE
|
+-- Q8_0 465 MB
+-- BF16 862 MB
+-- F16 862 MB
|
+-- treat as:
    shard / adapter / metadata / non-main artifact
|
+-- do not assign as primary until runtime loads it

```

11 D-008: Endpoint and Port Layout

```

Endpoint Layout: Two-Endpoint Baseline
|
+-- brain.local
| |
| | +-- port:
| | | 8081
| |
| | +-- role:
| | | Brain
| |
| | +-- preferred profile:
| | | brain-primary-14b-gguf
| |
| | +-- experimental profile:
| | | brain-primary-27b-nvfp4-mtp
| |
| | +-- contract:
| | | json_taskpack
|
+-- coder.local
| |
| | +-- port:
| | | 8082
| |
| | +-- role:
| | | Coder
| |
| | +-- profile:
| | | coder-primary-12b-fable
| |
| | +-- default quant:
| | | Q4_K_M
| |
| | +-- contract:
| | | unified_diff

Endpoint Layout: Three-Lane Variant
|
+-- brain.local      :8081 -> Brain JSON
+-- coder.local      :8082 -> Coder diff
+-- reviewer.logical :8082 -> Reviewer JSON, logical lane only

Rule:
    logical_only = true means scheduler label, not separate hardware

```

12 D-009: Parser Contract Boundary

```

UNTRUSTED MODEL OUTPUT
|
+-- Brain response
| |
| | +-- maybe JSON
| | +-- maybe prose

```

```

|   +-- maybe accidental markdown
|
+-- Coders response
|   |
|   +-- maybe diff
|   +-- maybe explanation
|   +-- maybe invented files
|
+-- Reviewers response
|   |
|   +-- maybe review JSON
|   +-- maybe vague opinions
|
=====
PARSER CONTRACT BOUNDARY
=====

+-- Brain parser
|   |
|   +-- extract JSON
|   +-- validate schema
|   +-- validate graph node types
|   +-- validate completion gate
|
+-- Coders parser
|   |
|   +-- extract unified diff
|   +-- validate touched files
|   +-- git apply --check
|   +-- changed-line telemetry
|
+-- Reviewers parser
|   |
|   +-- extract review JSON
|   +-- validate risk fields
|   +-- policy decides warn/block/allow
|
ACCEPTED ARTIFACTS
|
+-- taskpack.json
+-- agent.graph.json
+-- extracted.patch
+-- review_result.json
+-- verify_result.json

ONLY ACCEPTED ARTIFACTS CAN MOVE THE GRAPH

```

13 D-010: WO-006-C Script Workflow

```

leaf_model_assignments.py
|
+-- summary
|   |
|   +-- print profiles
|   +-- print role
|   +-- print repo
|   +-- print lanes
|   +-- print contracts
|
+-- routes
|   |
|   +-- all
|   +-- stable
|   +-- quality
|   +-- low-memory
|   +-- repair
|
+-- write-json
|   |

```

```

|   +-- output:
|       models/model-assignments.json
|
+-- doctor
|
|   +-- input:
|       model-library.md
|
|   +-- check:
|       expected model repos
|       expected quant labels
|       expected lanes
|
|   +-- exit codes:
|       0 = all anchors found
|       1 = anchors missing
|       2 = documentation missing

```

14 D-011: Documentation Doctor Tree

```

doctor --doc model-library.md
|
+-- does file exist?
|   |
|   +-- no
|   |   |
|   |   +-- exit 2
|   |
|   +-- yes
|   |   |
|   |   +-- read text
|   |
+-- required model anchors
|   |
|   +-- Brian6145/Qwen3.6-27B-Claude-Opus-Sonnet-Distilled-NVFP4-MTP
|   +-- tvall43/Qwen3.5-14B-A3B-Claude-4.6-Opus-Reasoning-Distilled-reap-gguf
|   +-- yuxinlu1/gemma-4-12B-coder-fable5-composer2.5-v1-GGUF
|   |
+-- required quant anchors
|   |
|   +-- Q2_K
|   +-- Q3_K_M
|   +-- Q4_K_M
|   +-- Q6_K
|   +-- Q8_0
|   |
+-- required lane anchors
|   |
|   +-- slot.brain.plan
|   +-- slot.coder.patch
|   +-- slot.review.scope
|
RESULT
|
+-- if all found:
|   exit 0
|
+-- if any missing:
|   exit 1

```

15 D-012: Performance Prediction and Benchmark Gate

```

Predicted Performance Page
|
+-- planning estimates
|   |

```

```

|   +--- 4070 Brain 14B + 3060 Fable 12B:
|       |   60-70 tk/s aggregate target
|       |
|   +--- 27B Brain + Fable 12B:
|       |   45-70 tk/s experimental
|
+--- overclock estimate
|   |
|   +--- mild tune:
|       |   +3% to +8%
|       |
|   +--- good cooling + tuned power:
|       |   +5% to +12%
|       |
|   +--- unstable overclock:
|       |   no stable profile
|
+--- benchmark gate
|   |
|   +--- run 3 cold starts
|   +--- run 3 warm starts
|   +--- run short prompts
|   +--- run long prompts
|   +--- record tk/s
|   +--- record TTFT
|   +--- record VRAM
|   +--- record thermals
|   +--- record failures
|   +--- pass parser-contract eval

```

STABLE PROFILE CRITERIA

```

|
+--- no crashes
+--- repeatable tk/s within +/-15%
+--- parser contract acceptable
+--- no driver resets
+--- enough VRAM headroom

```

16 D-013: Edge Cluster Helper Scaling

Edge Cluster Role

```

|
+--- good tasks
|   |
|   +--- file classification
|   +--- log summarization
|   +--- symbolic training export
|   +--- test result clustering
|   +--- README section detection
|   +--- completion-map inspection
|
+--- poor tasks
|   |
|   +--- one giant shared autoregressive decode
|   +--- single long patch
|   +--- large Brain reasoning pass

```

Scaling estimate

```

|
+--- simple independent-task model:
|    $T_{\text{cluster}} = T_{\text{single}} * N * 0.98$ 
|
+--- conservative coordinated model:
|    $T_{\text{cluster}} = T_{\text{single}} * N * 0.98^{(N-1)}$ 

```

Simple 0.98 multiplier

```

|
+--- 1 node  -> 1.00x
+--- 2 nodes -> 1.96x

```

```

+-- 4 nodes -> 3.92x
+-- 8 nodes -> 7.84x
+-- 16 nodes -> 15.68x
+-- 32 nodes -> 31.36x
+-- 64 nodes -> 62.72x

LeafOS use
|
+-- Hailo / edge accelerator:
|   classifier/helper/symbol export
|
+-- Jetson / Orin node:
|   light helper / summarizer / file map
|
+-- mini-PC + GPU:
|   remote coder or reviewer endpoint

```

17 D-014: Artifact Export Package Tree

```

W0-006-C Export Package
|
+-- documentation
|   |
|   +-- model-library.md
|   +-- model-library-real-assignments.md
|   +-- model-library-with-performance.md
|
+-- work order
|   |
|   +-- W0-006-C-model-assignment-library-script.md
|
+-- script
|   |
|   +-- leaf_model_assignments.py
|
+-- generated output
|   |
|   +-- models/model-assignments.json
|
+-- supporting docs
|   |
|   +-- model_assignment_ascii_diagrams.tex
|   +-- model_assignment_diagrams.md
|   +-- README.md

Export Gate
|
+-- include:
|   docs
|   script
|   generated JSON
|   diagrams
|
+-- exclude:
|   __pycache__/
|   .pytest_cache/
|   .env
|   secrets
|   downloaded model weights

```

18 D-015: Failure and Demotion Tree

```

Model route attempt
|
+-- call model endpoint
|

```

```

+-- response received?
|
|   +-- no
|   |
|   |   +-- failure:
|   |   |   model_endpoint_down
|   |   +-- action:
|   |   |   use fallback route
|   |
|
+-- parser accepts contract?
|
|   +-- no
|   |
|   |   +-- failure:
|   |   |   parse_failure / schema_failure / prose_only
|   |   +-- action:
|   |   |   retry once with stricter header
|   |
|
+-- scope valid?
|
|   +-- no
|   |
|   |   +-- failure:
|   |   |   scope_failure / invented_file
|   |   +-- action:
|   |   |   reject patch, request repair
|   |
|
+-- git apply --check passes?
|
|   +-- no
|   |
|   |   +-- failure:
|   |   |   patch_apply_failure
|   |   +-- action:
|   |   |   route to repair
|   |
|
+-- tests pass?
|
|   +-- no
|   |
|   |   +-- failure:
|   |   |   verification_failure
|   |   +-- action:
|   |   |   repair route
|   |
|
+-- repeated failure?
|
|   +-- yes
|   |
|   |   +-- demote model profile for this run
|   |   +-- select fallback route
|   |
|   +-- no
|   |
|   |   +-- mark accepted
|   |   +-- update event log
|   |   +-- continue graph
|
DEMOTION POLICY
|
+-- 1 failure:
|   retry with stricter prompt
|
+-- 2 failures:
|   alternate quant or helper route
|
+-- 3 failures:
|   demote profile for run
|   require human review

```

19 Final Notes

The model assignment layer is only useful if it stays strict.

Fast model + invalid artifact	= failure
Slow model + valid artifact	= useful
Parser-valid output	= currency
Raw token speed	= only a routing hint